

SAIPAN, Northern Mariana Islands

WMO: 912320

Lat: 15.117N

Lon: 145.717E

Elev: 215

StdP: 14.58

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 41418

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	73.4	74.8	64.2	90.6	77.6	66.0	96.6	78.2	27.5	79.0	25.6	80.0	9.9	10	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	8.7	89.8	79.9	88.8	79.6	88.1	79.3	81.9	86.5	81.3	86.0	80.8	85.6	12.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.6	160.3	84.3	79.9	156.3	83.9	79.2	152.7	83.6	46.0	87.4	45.3	86.2	44.7	86.3	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.7	22.0	20.4	DB	71.0	93.0	2.4	3.8	69.2	95.7	67.8	97.9	66.5	100.0	64.7	102.8	(4)
(5)			WB	65.9	83.4	1.9	2.1	64.5	84.9	63.4	86.2	62.4	87.4	61.1	88.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.9	80.1	79.8	80.5	81.9	83.0	83.6	82.8	82.5	82.2	82.0	82.2	81.4
	DBStd	2.04	1.76	1.59	1.68	1.53	1.29	1.67	1.98	2.00	1.82	1.80	1.59	1.57
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11626	933	833	946	956	1023	1009	1018	1008	967	993	966	974
	CDD65	6151	468	413	481	506	558	559	553	543	517	528	516	509
	CDH74	65050	4156	3535	4433	5295	6326	6694	6363	6100	5668	5709	5583	5188
	CDH80	18273	736	593	938	1428	2042	2493	2144	1913	1667	1599	1549	1171
Wind	WSAvg	11.5	13.4	13.9	13.6	12.8	11.1	10.6	9.2	8.5	8.3	10.1	13.1	13.9
Precipitation	PrecAvg	78.3	3.9	3.4	2.7	3.1	3.1	4.9	9.6	12.8	11.5	11.0	7.5	5.1
	PrecMax	96.0	8.1	9.4	6.5	10.0	6.5	18.4	19.9	28.5	17.5	18.7	13.2	12.7
	PrecMin	46.7	1.1	0.9	0.4	0.6	1.1	0.8	5.5	6.9	5.6	6.2	1.5	1.7
	PrecStd	10.7	2.2	2.1	1.4	2.0	1.4	3.4	3.3	5.4	3.0	3.6	3.2	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.2	87.2	87.9	88.8	89.7	91.5	91.3	91.9	89.7	88.5	88.3	87.8
		MCWB	78.1	77.4	77.5	78.2	79.0	80.4	80.7	81.4	80.8	80.3	79.6	78.4
	2%	DB	86.1	85.5	86.4	87.8	88.2	89.6	89.1	88.7	88.2	87.9	87.5	86.4
		MCWB	76.9	76.5	76.9	77.7	78.5	79.5	80.1	80.4	80.1	80.1	79.4	78.0
	5%	DB	84.8	84.5	85.5	86.7	87.7	88.6	88.0	87.7	87.5	86.9	86.4	85.8
		MCWB	76.4	76.1	76.5	77.6	78.4	79.4	79.9	80.1	80.0	79.9	79.1	77.8
	10%	DB	84.0	83.7	84.5	85.9	86.8	87.8	87.1	86.5	86.3	86.0	85.8	84.7
		MCWB	76.1	75.7	76.0	77.4	78.3	79.2	79.7	79.8	79.7	79.6	78.9	77.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.8	79.8	79.4	80.7	81.4	81.8	82.8	82.3	82.4	82.1	81.9	81.0
		MCDB	85.0	83.7	85.1	85.5	86.5	87.8	86.7	87.7	87.1	86.0	85.8	84.5
	2%	WB	78.8	78.7	78.2	79.5	80.3	80.9	81.7	81.4	81.4	81.0	80.9	79.9
		MCDB	83.3	82.9	83.9	84.8	85.8	87.0	86.1	86.5	86.0	85.1	84.9	83.7
	5%	WB	78.0	77.8	77.5	78.8	79.6	80.2	80.9	80.8	80.8	80.6	80.1	79.1
		MCDB	82.5	82.0	83.0	84.4	85.1	86.2	85.7	85.6	85.3	84.8	84.2	83.0
	10%	WB	77.3	77.0	76.9	78.1	79.0	79.7	80.2	80.2	80.1	80.0	79.5	78.4
		MCDB	81.6	81.2	82.3	83.7	84.5	85.7	85.1	84.7	84.4	84.0	83.6	82.5
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.1	8.4	8.8	8.5	8.7	8.3	8.3	8.4	8.0	7.7	7.4
		MCDBR	9.2	9.1	9.3	9.7	9.2	9.5	9.1	9.3	9.5	8.7	8.4	8.2
		MCWBR	4.6	4.6	4.2	4.3	3.9	4.1	4.3	4.6	4.7	4.6	4.4	4.3
	5% WB	MCDBR	8.4	8.3	8.5	8.8	8.5	9.1	8.5	8.7	8.8	8.2	7.7	7.2
		MCWBR	4.7	4.8	4.4	4.5	4.3	4.3	4.5	4.8	4.7	4.7	4.6	4.4
Clear-Sky Solar Irradiance	taub	0.407	0.410	0.406	0.410	0.410	0.411	0.406	0.414	0.405	0.428	0.432	0.418	
	taud	2.485	2.470	2.495	2.515	2.537	2.537	2.566	2.538	2.570	2.489	2.423	2.449	
	Ebn at Noon	281	285	288	285	281	278	279	280	285	277	271	274	
	Edh at Noon	34	36	36	35	33	33	32	34	32	34	36	34	
All-Sky Solar Radiation	RadAvg	1534	1757	2016	2184	2193	2116	1862	1665	1673	1610	1621	1527	
	RadStd	86	89	79	95	77	149	139	156	107	154	84	66	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+196	+196
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page